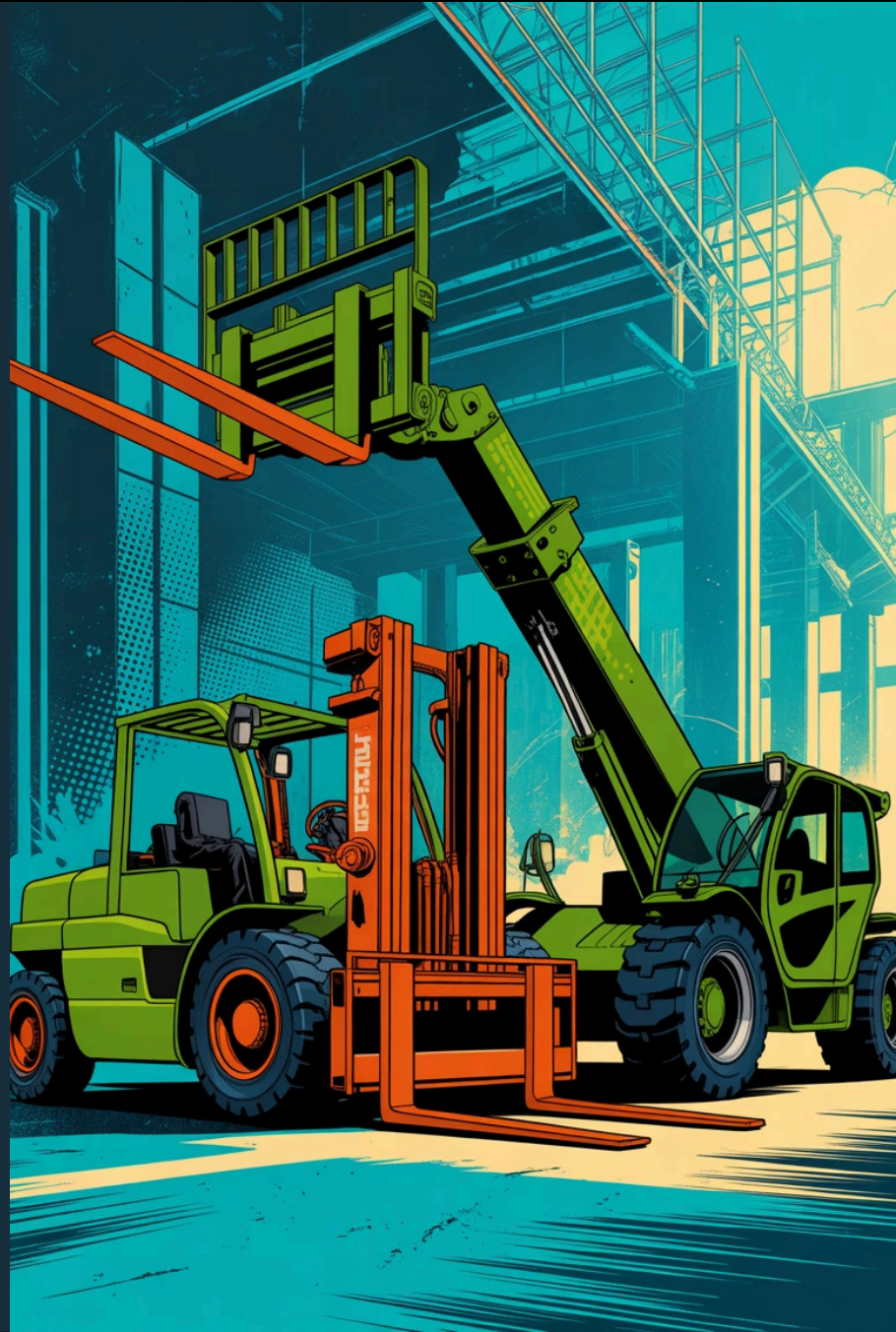
Forklift (Classes 1–5) vs Telehandler (Class 7)

Operator Training Comparison

Aligned to Genie Lift Pro Telehandler Trainer's Guide — Sections 1–8



SECTION 1: INTRODUCTION

Overview — Equipment Characteristics

Classes 1–5 Forklifts

Powered industrial trucks (PITs) are designed to lift, carry, and stack loads on firm, level warehouse or industrial floors. These machines use rear-wheel steering, which creates tail swing that operators must manage carefully. The vertical mast assembly blocks forward visibility when elevated, requiring operators to plan their moves and sometimes travel in reverse.

Capacity is determined by the data plate rating at a specified load center distance, typically 24 inches. Operators must verify that loads fall within these parameters before lifting.

Class 7 Telehandlers

Telehandlers are variable-reach rough-terrain forklifts designed for outdoor construction and industrial applications. Unlike standard forklifts, they feature a telescoping boom that extends forward and upward, dramatically changing the machine's center of gravity and stability characteristics.

These machines require level setup on firm ground. Load capacity is governed by complex charts that account for boom height, extension, and angle. A spotter is typically required during reach operations to maintain safety and coordinate placement.

Training Information Requirements

Classes 1–5 Forklifts

Operators must complete formal instruction combining classroom theory, hands-on practice, and performance evaluation before operating independently. Training must be specific to the truck type and workplace conditions.

Refresher training is mandatory after any incident, observation of unsafe operation, assignment to a new truck type, or significant changes to the workplace. Employers must maintain comprehensive training records documenting all instruction and evaluations.

When attachments are used, operators require additional attachment-specific instruction covering proper mounting, capacity changes, and safe operation techniques.

Class 7 Telehandlers

Training follows similar structure with formal instruction, practical exercises, and evaluation. However, the complexity of telehandler operations requires additional emphasis on load chart interpretation and boom dynamics.

The operator's manual and all applicable load charts must be stored in the cab and readily accessible. **Refresher training is required** after incidents, unsafe acts, new attachment installations, or worksite changes that affect operation.

Each attachment requires its own specific instruction module covering installation procedures, capacity implications, control functions, and operational techniques unique to that attachment type.

Safety Decals and Legend

Classes 1–5 Forklifts

Standard safety hierarchy uses four signal words: DANGER indicates immediate hazards that will cause severe injury or death. WARNING identifies hazards that could cause severe injury or death. CAUTION warns of hazards that may cause minor injury. **NOTICE** addresses property damage risks.

All safety decals must remain legible throughout the machine's service life. Replace damaged, faded, or missing decals immediately. Never paint over, remove, or obscure safety decals. These markings are critical safety devices, not decorative elements.

Class 7 Telehandlers

Telehandlers use the same four-level safety hierarchy. However, their decal systems are more extensive due to operational complexity. Critical decals show boom angle indicators, reach zone boundaries, capacity warnings at various configurations, and tip-over hazard zones.

Before each shift, operators must verify that all decals are present and readable from the operator's station. Maintenance staff must replace any damaged or illegible decals before returning the machine to service. Operators should read and understand all decal warnings before beginning operation, especially when working with new attachments.

Telehandler Features and Safety Equipment

Classes 1–5

Standard mast forklifts are not applicable to this section. Their feature set focuses on vertical lift capacity.

Typical safety equipment includes:

- Load backrest to prevent rearward shift
- Overhead guard protecting operator from falling objects
- Tilt mechanism for load stability
- Side-shift carriage (when equipped)

Class 7 Telehandlers

Telehandlers incorporate significantly more safety systems due to their variable geometry and rough-terrain operation:

- **Telescoping boom** with extension and retraction controls
- **Frame sway** system for lateral leveling on uneven ground
- **Stabilizers** that extend the stability footprint during reach operations
- **Rear axle lock** preventing oscillation during lifting
- **Multiple steering modes** (front-wheel, four-wheel, and crab steering) for maneuverability
- **Boom angle and extension indicators** showing real-time position relative to capacity charts
- **Overload warning systems** with visual and audible alarms alerting operators before exceeding safe limits

SECTION 2: REGULATIONS AND OPERATOR TRAINING

Overview of Training Requirements

Classes 1–5 Forklifts

Standard forklifts operate with a vertical mast and forks on firm, level floors. Rated capacity is determined by the data plate specification at a defined load center distance. The stability triangle concept governs safe operation, with the combined center of gravity of machine plus load remaining within the triangular footprint formed by the two front wheels and the pivot point of the steer axle.

Rear-wheel steering requires operators to manage tail swing during turns. When traveling, operators must keep forks low (4–6 inches) and tilt the mast slightly back to maintain stability and forward visibility.

Class 7 Telehandlers

The telescoping boom fundamentally changes capacity and stability calculations. As boom height and reach increase, available capacity decreases dramatically. Operators must use the correct load chart matching their specific machine model and installed attachment—never approximate or use a chart from a different configuration.

Critical operational sequence: Position the machine first with the boom retracted. Set the parking brake. Raise the load with the boom retracted. Extend the boom only when ready to place the load. Retract the boom before traveling. A dedicated spotter is required whenever the operator's line of sight is limited or when working near personnel, structures, or power lines.

Safe Use Plan Requirements

Both equipment classes require comprehensive safe use planning, though the specific elements differ based on operational characteristics and hazard profiles.

Classes 1–5 Forklifts

Safe use plans for standard forklifts address the fundamentals of materials handling on prepared surfaces. The plan must specify how operators will verify load weights and centers of gravity before lifting. It should define traffic patterns, pedestrian protections, and right-of-way rules.

The stability triangle remains the core concept—operators must understand how load position, mast tilt, and travel speed affect the combined center of gravity. Rear-wheel steering characteristics and tail swing management must be addressed. Travel protocols should mandate low fork height and slight mast-back tilt for optimal visibility and stability.

Class 7 Telehandlers

Telehandler safe use plans must address the boom's variable geometry and its dramatic impact on machine stability. The plan should specify how operators will select and verify the correct load chart for their machine-attachment combination. Procedures must cover how capacity varies with boom height, extension distance, and angle.

The operational sequence—position, brake, raise retracted, extend to place, retract to travel—should be explicitly documented. Plans must define when spotters are required and establish communication protocols. Ground conditions, slope limitations, and weather restrictions (particularly wind thresholds) should be clearly stated. Stabilizer and rear-axle lock deployment criteria complete the plan.

OSHA 1910.178 (l) — Operator Training Standard

OSHA's powered industrial truck standard establishes minimum training requirements for all equipment classes. While the regulatory framework is identical, application differs based on equipment characteristics.

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Classes 1–5 Forklift Compliance

Training must cover truck-related topics including operating instructions, capacity and stability principles, and vehicle inspection procedures. Workplace-related topics must address surface conditions, load manipulation, pedestrian traffic, and hazardous locations if applicable.

Practical training demonstrates the operator's ability to safely operate the specific truck type in the actual workplace environment. Evaluation confirms competency before independent operation is authorized. Refresher training is required at least every three years and more frequently if incidents, near-misses, or unsafe acts occur.

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Class 7 Telehandler Compliance

The same OSHA standard applies, but training content must address telehandler-specific topics including boom dynamics, load chart interpretation, and the critical importance of the position-raise-extend-retract sequence. Capacity variation with boom configuration requires extensive practical exercises.

Workplace-related training must cover rough-terrain considerations including ground bearing capacity, slope limitations, and wind effects on extended booms. Spotter communication protocols and exclusion zone management are essential topics. Each attachment type requires supplementary training documented as part of the operator's qualification record.

CSA B335 — Canadian Standards

Classes 1–5 Forklifts

The Canadian Standards Association B335 standard parallels OSHA requirements for forklift operator training in Canadian jurisdictions. Training must address the stability triangle, load center calculations, and the effects of rear-wheel steering on machine behavior.

CSA B335 emphasizes the importance of understanding how load position affects the combined center of gravity. Operators must demonstrate competency in load assessment, approach techniques, and travel procedures. The standard requires low fork height and controlled mast tilt during travel to maintain visibility and prevent tip-over.

Documentation requirements under CSA B335 mandate records of initial training, periodic evaluations, and refresher sessions following incidents or equipment changes.

Class 7 Telehandlers

CSA B335 recognizes the unique hazards of variable-reach rough-terrain forklifts. Training must thoroughly cover boom extension effects on capacity and stability. Operators must prove they can correctly select and interpret load charts for their specific machine and attachment configuration.

The standard requires training on the complete operational cycle: positioning on level ground, setting brakes, raising with boom retracted, extending only to place loads, and retracting before travel. Operators must understand when rear-axle locks and stabilizers are required and how to deploy them correctly.

Spotter requirements receive significant emphasis, with training covering communication methods, exclusion zones, and coordination during complex lifts. Wind restrictions and load sail area calculations are also addressed.

SECTION 3: FIVE PRINCIPLES OF SAFE OPERATION

Introduction to Five Principles

Safe operation of powered industrial trucks and telehandlers rests on five fundamental principles that protect operators, co-workers, and property. While these principles apply universally, their specific application varies between equipment types.

Avoid Hazardous Situations

Recognize and prevent dangerous conditions before they develop

Understand Capacity Limitations

Never exceed rated load limits for any configuration

Maintain Situational Awareness

Know your environment, equipment state, and nearby personnel

Operate Only Serviceable Equipment

Inspect before use and tag out defective machines

Protect All Personnel

Use spotters, barriers, and communication to prevent injuries

The following sections break down how each principle applies to standard forklifts versus telehandlers, highlighting similarities and critical differences.

Avoid Hazardous Situations

Classes 1–5 Forklifts

Standard forklifts operate primarily on prepared surfaces where hazards are more predictable. The vertical mast and fixed geometry create consistent stability characteristics that operators can learn and apply reliably.

Key hazards include pedestrian traffic in shared aisles, dock edges without adequate guards, overhead obstructions that could contact the mast or overhead guard, and floor conditions such as debris, spills, or damaged surfaces. The data plate provides clear capacity limits at the specified load center—exceeding these parameters creates immediate tip-over risk.

The stability triangle concept helps operators visualize safe operating limits. As long as the combined center of gravity stays within the triangle, the truck remains stable. Traveling with elevated loads or high mast positions moves the center of gravity dangerously forward and upward.

Class 7 Telehandlers

Telehandlers face significantly more variable hazards due to rough-terrain operation and boom extension. Ground bearing capacity must be verified before setup—soft soil can cause machine settling that changes boom angle and shifts the center of gravity during lifting.

Load charts define safe operating zones that change continuously as the boom moves. Even small changes in boom angle or extension can dramatically reduce available capacity. Operators must monitor angle and extension indicators constantly and understand how to interpret chart boundaries.

Wind loading on extended booms and suspended loads creates additional dynamic forces that reduce effective capacity. Site policies should define wind speed thresholds above which operations cease. Electrical hazards from overhead power lines are more significant due to boom reach—maintain approach distances with a dedicated spotter watching clearances. Barricade ground-level exclusion zones to keep personnel away from tip-over danger areas.

Safety Signs and Signal Words

Both Equipment Classes

All powered industrial equipment uses a standardized four-level safety labeling system. **DANGER** signals imminent hazards that will result in severe injury or death if not avoided. **WARNING** indicates potentially hazardous situations that could result in severe injury or death.

CAUTION identifies hazards that may result in minor or moderate injury. **NOTICE** addresses situations that could result in property damage but not personal injury.

Operators must respect these signal words as critical safety information, not suggestions. Follow the instructions on every safety decal without exception.

Classes 1–5 Specifics

Forklift safety decals focus on overhead clearances, capacity limits at specific load centers, tail swing hazards, and elevated load warnings. Operators must maintain overhead clearances to prevent mast contact with structures, sprinklers, or lighting.

Keep all decals visible and intact. Faded, damaged, or missing decals must be replaced immediately—they are required safety devices, not optional markings.

Class 7 Specifics

Telehandler decals are more extensive, covering boom angle zones, extension limits, capacity at various configurations, stabilizer requirements, and tip-over hazard areas. Zone markings on the boom indicate capacity regions—operators must know which zone they're operating in at all times.

Never operate a telehandler with missing or illegible decals, especially capacity charts and zone indicators. Without these references, there is no way to verify safe operation. The consequences of operating outside chart limits include catastrophic tip-over.

Hazard Classification

Understanding hazard categories allows operators to apply appropriate controls and personal protective equipment. Both equipment types share common hazards but face them with different frequencies and severity levels.

Tip-Over Hazards

Forklifts: Occur from overloading, elevated travel, or rapid turns with loads.

Telehandlers: More frequent due to boom extension moving center of gravity forward. Wind loading on extended booms increases risk.

Crush Hazards

Forklifts: Tail swing, pinch points between carriage and mast, pedestrians in travel path.

Telehandlers: All forklift risks plus exclusion zones beneath boom reach, stabilizer deployment areas, and load swing radius.

Electrical Hazards

Forklifts: Limited by mast height, but still require minimum approach distances to power lines.

Telehandlers: Significantly elevated risk due to boom reach bringing machine contact points close to overhead conductors. Spotters mandatory near power lines.

Fire Hazards

Both: Fuel systems, hydraulic fluid under pressure, electrical components, and hot exhaust. Battery charging areas require special ventilation and controls.

Match identified hazards to engineering controls (guards, interlocks), administrative controls (procedures, training), and personal protective equipment (hard hats, safety glasses, high-visibility vests). Site-specific plans should include electrical approach distances and wind thresholds when operating telehandlers. Establish barricades and exclusion zones to protect ground personnel during reach operations.

Work Area Safety

Classes 1–5 Forklifts

Warehouse and industrial environments require clearly marked pedestrian aisles with right-of-way rules favoring foot traffic. At intersections and blind corners, operators must sound the horn and slow to a safe speed that allows complete stops within the visible distance.

Loading dock edges must have guards or barriers to prevent trucks from driving off. Adequate lighting allows operators to see pedestrians, obstacles, and floor hazards. Maintain floor surfaces free of debris, holes, or damage that could cause loss of control or load instability.

When operating in areas with pedestrian traffic, reduce speed and maintain heightened awareness. Use mirrors to monitor blind spots, especially when reversing. Never assume pedestrians see or hear the truck approaching.

Class 7 Telehandlers

Rough-terrain worksites present more variable hazards. Before beginning lift operations, barricade the placement zone to exclude all unauthorized personnel. Use cones, barriers, or caution tape to establish a clear perimeter. Post warning signs indicating overhead work in progress.

Assign a dedicated spotter to watch for approaching personnel, monitor boom clearances, and coordinate load placement. Establish clear hand signals or radio communication protocols before starting work. The spotter maintains exclusive focus on safety—they perform no other tasks during lift operations.

Verify minimum approach distances to overhead power lines and treat all lines as energized. Even de-energized lines can be re-energized without warning. Monitor wind conditions continuously—cease operations if wind speeds exceed site policy thresholds. Verify ground bearing capacity supports the loaded machine weight—soft or sloped ground requires alternative positioning or outrigger mats.

Loading — Capacity and Stability

Classes 1–5 Forklifts

Before lifting any load, verify its weight using shipping documentation, warehouse management systems, or scales. Confirm the load center distance—the horizontal distance from the fork face to the load's center of gravity. Standard forklift data plates assume a 24-inch load center, but actual loads may differ.

Center the load on the forks so weight distributes evenly across both fork blades. Tilt the mast back to secure the load against the backrest and improve stability during travel. Never exceed the rated capacity shown on the data plate—capacity decreases if the load center exceeds the rated distance.

When stacking loads, verify that lower loads can support upper loads without crushing. Place heavier loads on bottom positions. Ensure stacks remain stable and won't topple from vibration or minor impacts.

Class 7 Telehandlers

Load chart verification is critical and complex. Select the chart that exactly matches your machine model and installed attachment—using an incorrect chart risks catastrophic overload. Charts show capacity varying by boom height, extension, and angle. Some charts also vary by stabilizer position and load zone.

Confirm the actual load weight including all rigging hardware, slings, spreader bars, and the attachment itself. Account for sail area—large flat loads catch wind that adds dynamic loading. In gusty conditions, reduce working capacity to maintain safety margin.

Always set the parking brake before raising loads. Raise with the boom fully retracted to maximize available capacity. Extend the boom only when positioned correctly to place the load. Never attempt to reposition the machine with the boom extended or load elevated—retract and lower first, then move.

Raising Loads Safely

Classes 1–5 Forklifts

Raise loads only when the truck is completely stopped and properly positioned. Never attempt to lift while moving—motion adds dynamic forces that can cause tip-over or load shift. During travel, keep the load at minimal elevation (4–6 inches) to maintain stability and visibility.

Avoid abrupt mast motions that create shock loading. Smooth hydraulic control prevents load bouncing and reduces stress on the lifting mechanism. When raising to maximum height, watch for overhead obstructions including structures, sprinkler heads, lighting fixtures, and door frames.

Before raising, scan the area above and around the intended lift path. Confirm adequate clearance for the load at its highest point. Consider mast tilt angle—tilting back too far at full height can bring the load close to the overhead guard or other rear obstacles.

Class 7 Telehandlers

The golden rule for telehandlers: **raise with the boom retracted, extend only to place**. This sequence maintains maximum available capacity during the lift and minimizes forward center of gravity shift. Extending the boom while elevated dramatically reduces capacity and creates severe tip-over risk.

Set the parking brake before initiating any boom raise operation. On machines with rear-axle locks, verify the lock engages per the indicator. Keep the load as close to the machine as possible during raising—even small forward shifts in center of gravity can exceed chart limits.

Never attempt to reposition the machine while the boom is elevated. If the placement position is incorrect, retract the boom, lower the load, and reposition the machine. Use a spotter to coordinate final placement—the operator cannot see the landing zone when the boom is extended and elevated.

Traveling With Loads

Forklifts: Low and Back

Maintain fork height at 4–6 inches above the ground surface during all travel. Tilt the mast slightly back to secure the load and move the center of gravity rearward, improving stability.

If forward visibility is blocked by the load, travel in reverse with the load trailing behind. Use mirrors and watch for pedestrians and obstacles. Sound the horn at intersections, doorways, and blind corners.

No Turns on Ramps

When ascending or descending ramps, keep the load on the uphill side—load-end upgrade whether traveling forward or reverse. Never turn on a ramp—complete all turns on level surfaces before entering the slope.

Reduce speed on ramps and maintain smooth control inputs. Abrupt braking or acceleration on slopes can cause tip-over or load shift.

Telehandlers: Stowed and Slow

Boom must be fully retracted and lowered during all travel. Keep the load as low as safely possible—never travel with the boom elevated or extended. These configurations move the center of gravity dangerously forward and create massive tip-over risk.

Travel speeds must be slow and deliberate, especially on rough terrain. If a spotter assisted with placement, maintain spotter coordination during travel to clear obstacles and direct the path.

Tip-Over — Prevention and Response

Tip-over is one of the leading causes of forklift and telehandler fatalities. Understanding prevention and proper response can save lives.

Classes 1–5 Forklifts

Always wear your seatbelt. In a tip-over event, the seatbelt keeps you in the protected zone of the overhead guard. Stay in the operator compartment—do not attempt to jump clear. Brace your feet firmly on the floor, grip the steering wheel with both hands, and lean away from the direction of the fall. The overhead guard is designed to protect you if you remain inside.

Never jump from a tipping forklift. Operators who jump are frequently crushed by the overhead guard or the falling load. Your survival instinct will urge you to jump—override it. Stay in, hold on, brace.

Prevention is straightforward: respect capacity limits, travel with forks low, avoid elevated turns, maintain proper load positioning, and operate smoothly without abrupt inputs.

Class 7 Telehandlers

Seatbelt use is mandatory at all times. Telehandlers tip more easily due to boom extension shifting the center of gravity. The same survival response applies: stay in the cab, brace against the direction of fall, grip the steering wheel, plant your feet.

Boom extension dramatically increases tip-over risk. As the boom extends forward, the combined center of gravity of machine plus load moves beyond the stability footprint. Stabilizers and rear-axle locks expand the footprint somewhat, but longitudinal stability limits still govern—extending too far will tip the machine forward regardless of stabilizers.

Prevention requires strict adherence to load charts, proper operational sequence (position-raise-extend-retract), use of stabilizers per chart requirements, operation on level ground, and wind awareness. Never jump from a tipping telehandler.

Component Damage and Defects

Classes 1–5 Inspection Points

Daily pre-operation inspection must include fork condition (checking for cracks, bends, or wear), mast rails (looking for damage or misalignment), hydraulic hoses and fittings (checking for leaks or damage), and tires (verifying proper inflation and tread depth).

Immediately tag out and remove from service any truck with cracks in forks or mast, hydraulic fluid leaks, excessive tire wear, or any other defect that could affect safe operation. Do not attempt to operate defective equipment "just this once"—even minor defects can cause catastrophic failures.

After repairs are completed, qualified personnel must re-inspect the truck and authorize return to service before operators resume use.

Class 7 Inspection Points

Telehandler inspections add critical boom-specific components: boom extension and retraction mechanisms, boom pins and bushings, stabilizer cylinders and pins, rear-axle lock function, load charts present and legible, and all capacity/zone indicator systems operational.

Pay special attention to hydraulic hoses on the boom—they flex repeatedly during operation and can develop leaks or weak spots. Inspect stabilizer welds and mounting points for cracks. Verify that boom angle and extension indicators match actual positions.

Tag out any machine with hydraulic leaks, damaged boom components, missing or illegible load charts, non-functional indicators, or stabilizer defects. Re-inspection after repair is mandatory before returning to service.

Crush Hazards

Classes 1–5 Forklifts

Rear-wheel steering creates tail swing that can trap pedestrians or strike objects behind the operator's field of view. Always check clearances before turning, especially in tight spaces. Use mirrors to monitor the rear corners during turns.

Pinch points exist between the carriage and mast, between forks and pallets, and anywhere moving components can trap hands or body parts. Never reach into these areas while the truck is operating. Keep hands and feet inside the operator compartment.

Never allow anyone to stand or walk under elevated forks or loads. Even a small object falling from height can cause serious injury. **Never carry passengers on forks, carriage, or overhead guard**—these are not designed for personnel transport and offer no fall protection.

Class 7 Telehandlers

All forklift crush hazards apply, plus additional risks from boom reach operations. Establish and enforce exclusion zones beneath the boom and around the load path. No personnel should enter these zones during lifting operations.

Load swing during boom extension or retraction can strike personnel standing near the load path. Even in calm conditions, loads can swing during acceleration or deceleration of boom movements. In windy conditions, suspended loads swing unpredictably.

Stabilizer deployment presents crush hazards as outriggers extend—ensure the area is clear before deploying or retracting stabilizers. Maintain exclusion zones throughout reach operations. The spotter's primary responsibility is protecting personnel from crush hazards.

Explosion and Fire Hazards

Classes 1–5 Forklifts

Fuel-powered forklifts generate heat, sparks, and combustion byproducts. Operate in well-ventilated areas to prevent carbon monoxide accumulation. Never operate gasoline or LPG forklifts in enclosed spaces without adequate ventilation—carbon monoxide is colorless, odorless, and deadly.

Control ignition sources around flammable materials. Hot exhaust systems can ignite vapors or combustible dust. When operating in hazardous locations classified for explosive atmospheres, use only forklifts specifically rated and approved for that environment.

Follow proper fueling and charging procedures. Fuel only in designated areas away from ignition sources. Battery charging requires dedicated ventilated areas to disperse hydrogen gas generated during charging. Never smoke or use open flames near fueling or charging operations.

Class 7 Telehandlers

Diesel-powered telehandlers produce less carbon monoxide than gasoline engines but still require ventilation. Exhaust temperatures remain high enough to ignite flammable materials. Keep the machine clear of debris accumulation around the engine compartment and exhaust system.

Eliminate all ignition sources when working near flammable vapors or combustible dust. This includes not only the exhaust system but also electrical components that could spark. Battery-powered telehandlers eliminate combustion risks but introduce electrical